A Reproducible Audit of Name-Channel Noninterference in a Frozen Language Model

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Abstract

We audit structural name-channel isolation in a frozen Qwen2.5-1.5B derivative, extending an earlier generated-fixture study with 32 real CV excerpts, eight occupations, two job contexts per CV and four controlled header names. A frozen pre-inference plan specifies a 768-forward experiment, direct public-state comparisons and an equivalent external-canonicalization baseline. Original named inputs show a mean within-CV probability range of 9.75 percentage points; name swaps flip binary decisions in 9/64 contexts. The role-isolated scoring path has zero matched-name score variation; all 6,144 direct counterfactual public-stage comparisons and 128 native-evidence hidden/vocabulary comparisons match exactly. A category-pair relevance diagnostic is 62.5% for named scoring and 68.75% after isolation, but its paired bootstrap interval includes no improvement; it is not human hiring accuracy. Strong-baseline timing is approximately equal, with a median paired internal/external ratio of 1.002. Raw private embeddings still distinguish all four names, proving that decision-path invariance is not whole-model name erasure. An expanded 31-source review identifies close prior work and limits the contribution to a reproducible implementation and audit. No universal racial fairness, independent human validation or first-method claim is established.

1. Research question and contribution

Can a frozen language-model scorer be made exactly invariant to a designated applicant-name channel, while retaining its native computation on the same evidence? We distinguish this architectural question from two harder questions: whether every identity reference in arbitrary prose can be discovered, and whether the resulting hiring policy is substantively fair. The latter questions require evidence not supplied by name swaps alone.

This report extends an earlier synthetic implementation study with real CV and job excerpts, a strong equivalent external-canonicalization baseline, clustered analysis, and a wider novelty assessment. Its contribution is an auditable experiment and implementation artifact. It is not a claim to invent activation editing, anonymization or noninterference, and it does not establish whole-model racial-bias erasure. The author's single-runtime validation is distinct from independent human evaluation or replication by another research group.

2. Literature review and novelty assessment

Review method and boundaries

The following background synthesis distinguishes the prior synthetic release from the new real-excerpt experiment. This is a structured narrative review, not an exhaustive systematic review or a PRISMA study. Sources were checked on 3 November 2026 through publisher pages, author manuscripts and official documentation. The original 25 selected references cover six questions: why audit names, which fairness property is intended, whether behavior reveals internal information, what architectural exclusion guarantees, how to preserve token efficiency, and how to evaluate a defensible artifact. `search-log.json` records the search families, selection rules and exclusions. `references.json` records source locations, version caveats and how extensively each source was inspected. Abstract-only checks are identified rather than presented as complete paper replications. The bibliography is in `references.bib`.

The review distinguishes a prior paper's findings from our deductions. A citation motivates a design choice; it does not validate this implementation. We do not infer the absence of other work from this targeted search. No numerical outcomes from different models, tasks or populations are pooled into a meta-analysis.

2.1 Names as an experimental intervention in hiring

Bertrand and Mullainathan (2004) [6] establish a correspondence-audit precedent: vary names associated with social groups while controlling resume content and measure employer responses. This motivates matched-name interventions, not assigning a person's actual race from a name. Our synthetic model-score checks are not field evidence about employer callbacks.

Wilson and Caliskan (2024; corrected manuscript 2026) [7] audit resume retrieval using text-embedding similarity. Their first-page erratum says gender-only preferences were reversed by a coding error, while race-only and intersectional results were unaffected. We therefore avoid the uncorrected abstract's gender figures. Their retrieval setting differs from our autoregressive model readout; its results motivate auditing but cannot be imported as our baseline bias rate.

Armstrong et al. (2024) [8] study GPT resume assessment and generation, showing why evaluation should distinguish scoring existing evidence from producing new candidate descriptions. We study a deterministic restricted scoring interface, not generated recommendations or an end-to-end recruitment workflow. Together, these audits justify controlled identity swaps while leaving deployment-wide claims dependent on task, prompt, model and population.

2.2 Fairness is broader than an identity channel

Kusner et al. (2017) [3] formulate counterfactual fairness using a causal model. Our equivalence relation instead fixes retained evidence Q and varies designated identity text. We neither identify the causal effect of race nor hold constant every consequence of discrimination. A job history or address can remain a racial proxy even when applicant-name substitution cannot affect the score.

Dwork et al. (2012) [11] locate individual fairness in a task-relevant similarity metric. Our exact equality of evidence is a deliberately narrower relation; it does not solve how two different histories ought to be compared. Raghavan et al. (2020) [9] examine hiring vendors' validation claims, making job relevance and validation choices part of the research question rather than assumed outputs. Selbst et al. (2019) [10] warn against abstractions that exclude the surrounding social process. This means reviewer authority, applicant appeal, retained proxies and selection policy belong in our limitations, not outside the system boundary.

The defensible conclusion is conditional name-channel noninterference, not "truly unbiased hiring." Equal scores can be equally inaccurate. A harmful job rubric stays harmful when its name input is disconnected. Real deployment needs validity evidence, stakeholder review, accessible correction and subgroup audits in addition to this narrow technical property.

2.3 Behavioral equality does not establish representation erasure

Hewitt and Liang (2019) [15] show why probe controls and selectivity matter: probe success can partly reflect the probe's learning ability. Conversely, failure of one fitted probe does not certify that no decoder can recover a concept. Gonen and Goldberg (2019) [25] demonstrate residual gender structure after particular word-embedding debiasing procedures. That finding concerns those embeddings; we use it as a warning against generalizing one favorable bias metric.

Our simplest counterexample is $H(q,n)=(q,n)$, with output $S(H)=q$. Swapping n cannot change S, yet n is perfectly recoverable from H. Thus removing observable score bias does not imply removing activation-space information. Even information about names is not itself an unfair decision: knowing a name's linguistic origin is compatible with prohibiting its influence on a hiring score. The distinction is encoding versus causal use, under a stated intervention.

Amnesic probing (Elazar et al., 2021) [16] links representation interventions to behavioral changes, rather than treating a probe alone as an explanation. Our public-stage equality measurements similarly examine more than a final score, but only after the explicit evidence boundary. Raw source embeddings, model weights and any private diagnostic branch remain outside that erasure claim.

2.4 Erasure, adapters and structural routing are different interventions

INLP (Ravfogel et al., 2020) [13] repeatedly fits linear attribute predictors and projects representations into their nullspaces. LEACE (Belrose et al., 2023; manuscript v4, 2025) [14] derives a closed-form minimal-distortion linear eraser under a precise guardedness definition. Its convex-loss linear-prediction guarantee must not be restated as unrestricted independence or inability of every zero-one-loss threshold classifier to recover a label.

For example, balanced Z with X=-1 or +1 when Z=0, and X=0 when Z=1 has equal class means and zero covariance. Nevertheless, thresholding X>0 predicts Z=0 with 75% accuracy; a nonlinear decoder of |X| recovers Z exactly. This is our finite mathematical witness, not a replication or refutation of LEACE's stated theorem. It illustrates why the predictor family and loss must accompany the word "erasure." Sample-estimated erasers also require held-out leakage tests.

LoRA (Hu et al., 2022) [17] freezes base weights and learns low-rank updates. QLoRA (Dettmers et al., 2023) [18] combines frozen quantized weights with adapters and specific memory-saving techniques. Our earlier MLX affine-four-bit adapter experiments should be called quantized LoRA, not a faithful NF4/double-quantized QLoRA replication. Neither adapter efficiency method by itself provides a universal fairness guarantee. This final canonical-evidence release trains no adapter and does not change base parameters.

Concept bottleneck models (Koh et al., 2020) [19] and ReFT (Wu et al., 2024) [20] offer alternative controlled representations or interventions. They help frame future learned evidence extractors, but are not evaluated baselines here. The present tradeoff is explicit: deterministic isolation gives a conditional dependence guarantee while shifting semantic correctness to reviewed roles.

2.5 The closest conceptual foundation is information-flow security

Sabelfeld and Myers (2003) [12] explain end-to-end information flow; Chong et al. (2007) [4] distinguish confidentiality and integrity in an enforced system. These are closer foundations for our theorem than statistical debiasing. Let D~D' mean Q(D)=Q(D'). A score S=G(Q(D)) is constant on each equivalence class. The proof is ordinary deterministic composition, not a new fairness theorem. Authenticated reviews protect integrity of the chosen partition; they cannot prove that the partition faithfully captures applicant identity.

This framing yields an essential baseline. An external canonicalizer computing the identical Q followed by the same tokenizer, weights, positions and readout has the same score as our internal canonicalizer. Embedding the raw source first does not strengthen that output guarantee. It is a trace/design constraint and adds work; names still exist in the raw private activations. Position, timing, admission, error and audit channels need separate treatment. A correct public score theorem is not a claim that the entire computer leaks no identity.

Internal placement is therefore an implementation characteristic, not evidence of novelty or model unlearning. The artifact's contribution is an inspectable, tested integration of reviewed source roles, native evidence tokenization and frozen-model scoring, with explicit failure cases and reproducible records.

2.6 Tokenization efficiency and annotation reliability

Sennrich et al. (2016) [1] motivate subword segmentation for open-vocabulary NLP; the official Tokenizers pipeline [2] separates transformations that can affect boundaries. Our deduction is algebraic: a codec satisfying C(a+b)=C(a)||C(b) for every scalar split is forced to encode scalars independently. It generally cannot also preserve every merge of whole-string BPE. Joining reviewed evidence before one tokenizer call avoids this incompatible requirement.

The exact result is n(Q)=|T(Q)| transformer positions, not globally minimum tokens and not total compute parity with unrestricted input. Raw-source embedding and review verification remain overhead. The earlier synthetic release's measured 4.10x median paired speedup compares our two service packages; it is not a speedup over a strong optimized external-canonicalization baseline. That baseline was not timed in the earlier release; the new experiment below now measures it. It is also not an isolated ablation of token count because review APIs differ.

Artstein and Poesio (2008) [5] distinguish agreement measures and interpretation. Two credentials agreeing can share one error. Our synthetic oracle checks span mechanics, registered omissions and conflicting payloads, not human semantic accuracy. Independent adjudication must measure whole-document identity omissions and false evidence removals, including unknown aliases, transliteration, same-spelled skills and ambiguous third parties. Character agreement alone can look excellent while missing a short identifying span.

2.7 Evaluation, reporting and research gaps

CheckList (Ribeiro et al., 2020) [21] motivates targeted behavioral tests alongside aggregate accuracy. We separate name invariance, evidence sensitivity, annotation failure controls, public-stage equality, native-evidence parity and efficiency. The prior release's 108 controlled forwards repeat just three evidence bodies; its 3,360 array comparisons are correlated engineering checks, not independent fairness trials. Zero observed failures do not yield a population failure-rate estimate without a sampling model. The universal conditional argument comes from architecture, not extrapolation from those tests.

Dodge et al. (2019) [22] motivate reporting development choices and computation budgets. We preserve the source freeze and distinguish inherited GPU measurements from this later literature audit. Model cards (Mitchell et al., 2019) [23] and datasheets (Gebru et al., 2021) [24] motivate explicit intended use, limitations and data provenance. The earlier release's conformance corpus was generated; the supplied public CV/job datasets belonged to earlier project exploration and did not supply validation for that earlier extension. The new controlled study below now uses real excerpts without claiming human-validated hiring usefulness.

The prior review identified an independent real-document role study, a useful hiring-label benchmark and a competitive external-canonicalization baseline as missing evidence. The present experiment supplies controlled real-excerpt conformance and the strong external baseline; the human-label studies remain unfulfilled. Before a systems publication, the main missing evidence is a sharper threat model, implementation-level verification or independent audit, and broader workloads/hardware. The current artifact supports a transparent technical-report/preprint draft; novelty and venue suitability remain reviewer judgments.

Comparison of guarantees (our synthesis)

Approach	What its result concerns
Name-swap hiring audits [6-8]	Measured outcomes on specific prompts, names and tasks; no universal erasure theorem
INLP / LEACE [13-14]	Specified linear representational criteria; not all decoders or every unseen distribution
Quantized LoRA / ReFT [17-18,20]	Efficient adaptation/intervention; fairness requires additional objectives and evaluation
Concept bottlenecks [19]	Predictions through supervised concepts; concept correctness remains an assumption
Information-flow enforcement [4,12]	Dependence under a policy and trusted boundary; the policy can be socially inadequate
This artifact	Equal scoring-path tensors for equal reviewed Q; no automatic semantic discovery or whole-model erasure

Closest newer work and the revised novelty conclusion

Karvonen and Marks (2025) [26] directly mitigate hiring bias through affine activation edits, including more realistic contextual and proxy settings. Russinovich (2026) [27] already supplies a marked-span veto with exact outside activation noninterference under fixed geometry. Its canonical boundaries and length assumptions are explicit. These are material omissions from the initial review: we now reject broad first-method claims. Canonical evidence routing handles variable raw-name lengths by rebuilding its scoring sequence, but that functional property also follows from ordinary external canonicalization.

CaMeL (Debenedetti et al., 2025) [28] places capability-based flow enforcement around LLM agents. Tamkin et al. (2023) [29] audit demographic effects across decision scenarios; Marshall et al. (2024; revised 2025) [30] derive affine concept editing for activation steering; Hoffstedde et al. (2026) [31] test counterfactual names and mitigation in Japanese hiring. These sources support a narrow artifact/audit contribution, not a new unlearning algorithm. The companion novelty matrix records rejected priority claims and search queries.

3. Architecture and formal target

Let D contain exact source spans tagged as identity or retained evidence. Let X(D) concatenate all source text and Q(D) concatenate only retained spans. With the original tokenizer T, embeddings E and fixed scoring network G, execution is:

```
R(D) = E(T(X(D)))        # complete named source embedded first
H0(D) = E(T(Q(D)))       # evidence rebuilt inside the backend
H_l(D) = F_l(H_(l-1)(D)) # original 28 transformer blocks
S(D) = original_readout(H_L(D)).
```

R supplies no tensor, position, mask, cache or length to the evidence branch. Names enter original raw embeddings intact. Explicit code inside the backend constructs Q from reviewed roles; the model has not learned to discover every name autonomously. No adapter is trained in this release. Base parameters and checkpoint files are verified before and after inference.

For two accepted documents with Q(D)=Q(D'), deterministic tokenization gives identical evidence IDs. Equal embeddings, masks and positions propagate through each unchanged layer, giving equal scoring states and scores. The statement covers admitted unseen name values under the role/evidence premise. It does not cover undeclared body aliases, errors, admission, timing or private diagnostics.

An external canonicalizer that produces the same Q followed by the same G has the same output. Raw-source embedding first adds no mathematical score-isolation advantage. It is an implementation constraint with additional work, not a learned erasure mechanism. The transformer length is exactly |T(Q)|; total embedding visits include both |T(X)| and |T(Q)|. No globally minimum-token claim follows.

4. Frozen controlled study on real excerpts

The study plan and 12 inference/preparation source hashes were frozen before model inference. We sampled four CV records in each of eight categories from the provided Kaggle Resume Dataset, using deterministic hash ordering with seed

20261103. The source contains 2,484 records. We excluded the 24 original datathon case IDs, but do not certify disjointness from every prior local study or model pretraining. No fitting, adapter selection or outcome-dependent case selection was performed.

Each CV contributes two contexts: a category-matched job and a deterministically mismatched category's job, drawn from the cached 61-row LinkedIn listings sample. Categories are accountant, advocate, banking, engineering, healthcare, IT, sales and teacher. Two CVs per category use a plain instruction and two a selective instruction. Since styles use different CVs, their comparison is descriptive, not a causal within-CV estimate of selective-context effects.

Inputs contain the first 140 whitespace words of real CV text and the first 40 words of a real job description, plus job title and a fixed A/B relevance query in the original chat template. These are explicitly excerpt tasks, not full-CV validation. The four inserted header names are John Miller, Mohammed Hassan, Wei Chen and Neruvo Qaslin. The last is a constructed name, not certified absent from pretraining. Names are lexical interventions, not verified racial labels.

The inserted name span is exact by construction and authenticated using two temporary machine credentials. These are not human annotators. Original body names and other proxies are retained unchanged. The experiment therefore tests the declared header channel with real evidence, not automatic name discovery throughout natural CVs. CV/job text and keys remain local and are excluded from the public artifact; source hashes and selection code permit data-authorized reproduction.

For every context, we run the corrected path and original named-input scorer under all four names: 256 rows, 512 forwards. Each context also receives a native external-Q reference forward (64), comparing the entire final hidden array and full final-position vocabulary output. Corrected counterfactuals compare token IDs, embeddings, all 28 blocks, final normalization and full vocabulary arrays directly. There are 64 reference variants and 192 counterfactuals: 6,144 direct public-stage comparisons, not 6,144 independent applicants.

Warm timing uses one paired trial per CV on its matched job, after two warmups per method, with shuffled method order. Both methods verify the same signed role payload and model structure. Internal scoring additionally prepares and embeds the original source. The external baseline builds Q outside the backend and uses the native model. Neither timed path copies observed activation arrays. Signing and model loading are excluded. The run is sequential, not a concurrency or load-throughput benchmark. Timing adds 64 measured and 128 warmup forwards, for 768 actual forwards in total.

5. Results

All frozen-plan contexts completed without outcome-dependent exclusions. The checkpoint's original parameter digest and file hashes remained unchanged.

Measurement	Observed result
Real CV clusters / job contexts / name rows	32 / 64 / 256
Actual forwards including references, timing and warmups	768
Maximum corrected probability range across names	0, exact
Raw mean CV-level probability range	9.75 percentage points
Bootstrap interval for that mean	8.27 to 11.19 percentage points
Largest raw context probability range	34.33 percentage points
Raw versus corrected binary-flip contexts	9/64 versus 0/64
Direct counterfactual public-stage comparisons	6,144, all equal
Native-evidence final-hidden/full-vocabulary comparisons	128, all equal
Category-pair relevance: raw / corrected	62.5% / 68.75%
Paired relevance difference, bootstrap interval	+6.25 points, interval -0.78 to +14.06 points
Internal / external median service latency	303.5 / 299.6 ms
Median paired internal/external ratio	1.002
Pairwise distinct private raw-name states	4/4 names in every context

The raw range measures naming-cue sensitivity, not a measured racial group outcome disparity. A positive range alone does not identify racism or its mechanism. The corrected exact equality follows from the controlled evidence boundary and is checked on actual arrays. It is not extrapolated statistically from four known names to every unannotated name in arbitrary prose.

The median paired timing ratio has interquartile range 0.994-1.005. The strong external-Q baseline therefore receives no claim of speed inferiority. The earlier 4.10x speedup against scalar encoding must not be applied

to this comparison. These warm single-device timings are not a multi-device equivalence test or production throughput result.

6. Statistical interpretation and performance

The independent resampling unit is a CV, retaining its two jobs and four names together. We resample four CVs with replacement within each of the eight fixed categories, 5,000 times with a fixed seed, and report percentile intervals. These describe resampling stability in this small stratified dataset study, not worldwide population fairness. Repeated activation checks are not treated as independent Bernoulli trials. No universal failure-rate confidence bound is inferred from observing zero score gaps.

Category-pair relevance asks whether a CV's matched-category job receives a higher A/B margin than its mismatched job. Ties receive half credit. Each CV's raw result averages its four name variants. This is a narrow diagnostic based on source category labels; category matching is not human hiring accuracy or job-performance validity. Native evidence parity is tested separately through complete hidden/vocabulary arrays, not inferred from that diagnostic.

The observed paired relevance difference does not establish a population improvement: its bootstrap interval includes zero. No utility noninferiority margin was prespecified with domain experts, so we do not claim a formal hiring utility noninferiority result. We do establish native-evidence computational parity on the tested contexts. This includes the native model's mistakes.

7. Why score isolation is not whole-model erasure

For fixed Q, all four actual raw embedding states are pairwise distinct, while the scoring states coincide. Under the experiment's uniform four-name intervention distribution, the map from the name to raw state is injective. Consequently H(N|H_raw,Q)=0 and I(N;H_raw|Q)=log2(4)=2 bits. The invariant public state has I(N;H_public|Q)=0 for that same finite distribution. These statements concern this controlled name variable, not actual race, an open-world decoder or every private/timing channel. Distinctness uses direct array comparison, including shape, rather than hashes alone.

If original name embeddings remain distinct, an unchanged raw embedding stage cannot simultaneously be independent of the name. Later adapters cannot change the state that already existed. Thus whole-execution name erasure conflicts with the requirement to preserve that named raw stage. Keeping general name knowledge is compatible with isolating a hiring score from the identity channel.

A second boundary concerns performance claims. If two native named scores are a and b, any common corrected score c obeys

```
max(|c-a|, |c-b|) >= |a-b|/2.
```

This follows from the triangle inequality. Exact name invariance and exact preservation of unequal original scores are incompatible. Preserving task accuracy is a separate empirical goal; preserving native evidence computation is the exact target of this architecture.

The companion mathematical note also audits summed affine editing in [26]. For unit direction columns U and target b, a one-pass summed edit has residual (I-U^T U)(U^T h-b). It does not generally impose every target when directions are not orthogonal. Standard constrained least squares replaces that sum with U(U^T U)^(-1)(U^T h-b) for full-column-rank U. This known algebraic distinction does not refute [26]'s empirical results and supplies no unrestricted erasure guarantee. Exact rational examples are included in CPU tests.

8. Novelty, validity threats and publication status

The expanded search found direct activation-editing hiring research [26] and a closely related exact marked-span noninterference mechanism [27] omitted from the initial review. Accordingly we reject claims to be the first internal fairness intervention or first exact text-exclusion architecture. The limited contribution is an inspectable implementation and a controlled real-excerpt audit with a strong equivalent baseline and explicit negative result about whole-model erasure. Academic novelty and significance remain external reviewer judgments.

The main threats are small convenience coverage, eight occupations, four fixed names, repeated contexts, one pinned affine-four-bit Qwen derivative, one device, an A/B relevance readout, source truncation by declared excerpt policy and generator-known inserted roles. We do not evaluate free-form name detection, independent human role labels, real hiring outcomes, calibrated qualification scores, downstream shortlist fairness or every other proxy. Existing background names in Q can still influence outputs. Preserving Q under a header swap does not erase racial associations in qualifications or enforce a fair job policy.

The two credentials ensure mechanical payload agreement and integrity, not two independent judgments. No study participants, inter-annotator agreement or external peer-review approval are invented. The independent-review packet states what a separate evaluator must do. A new research release and polished citations do not establish acceptance or make this a deployment-ready hiring system.

9. Reproducibility and ethical handling

The release contains the pre-inference plan, source hashes, 256 public name rows, 64 native-reference records, paired timing records, clustered analysis, tests, data provenance, the novelty matrix and read-only verification. It preserves the previous releases and their failed or limited results. The model parameters and pinned files passed post-run integrity checks. Clef and other training jobs were not changed or interrupted. No base weights or private credentials are distributed. The text and code were developed with AI assistance.

Public data access does not eliminate privacy interests. Raw CV/job excerpts remain in an ignored local directory; public records contain only source hashes, categories, model outputs and the four artificial intervention names. Reproduction requires separately obtaining the source datasets and exact checkpoint. Cached job sampling limits representativeness and its redistribution terms are not assumed. Original source terms remain with the providers.

The code is suitable for inspection and reproducible research, not consequential automated hiring. Stronger claims require independent replication, adjudicated semantic roles, domain-expert job labels and a new prospective study. The present report supplies actual controlled real-data measurements and a documented novelty assessment, rather than treating those tasks as completed by assertion.

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